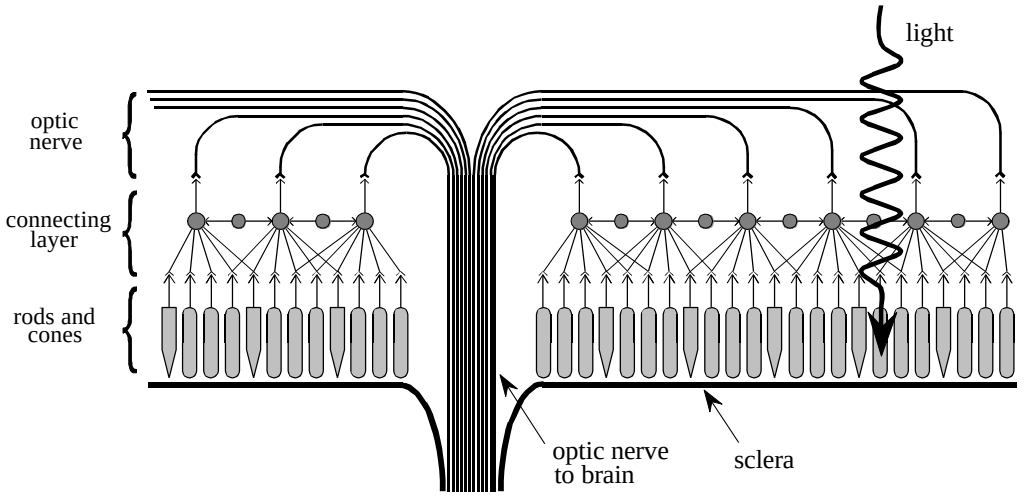

FIGURE 23-5
The human retina. The retina contains three principle layers: (1) the rod and cone light receptors, (2) an intermediate layer for data reduction and image processing, and (3) the optic nerve fibers that lead to the brain. The structure of these layers is seemingly *backward*, requiring light to pass through the other layers before reaching the light receptors.

Human vision overcomes the small size of the fovea by jerky eye movements called **saccades**. These abrupt motions allow the high resolution fovea to rapidly scan the field of vision for pertinent information. In addition, saccades present the rods and cones with a continually changing pattern of light. This is important because of the natural ability of the retina to adapt to changing levels of light intensity. In fact, if the eye is forced to remain fixed on the same scene, detail and color begin to fade in a few seconds.

The most common image sensor used in electronic cameras is the **charge coupled device** (**CCD**). The CCD is an integrated circuit that replaced most vacuum tube cameras in the 1980s, just as transistors replaced vacuum tube amplifiers twenty years before. The heart of the CCD is a thin wafer of

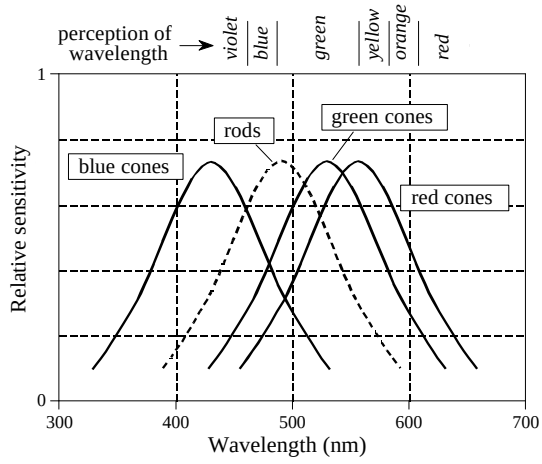

FIGURE 23-6
Spectral response of the eye. The three types of cones in the human eye respond to different sections of the optical spectrum, roughly corresponding to red, green, and blue. Combinations of these three form all colors that humans can perceive. The cones do not have enough sensitivity to be used in low-light environments, where the rods are used to detect the image. This is why colors are difficult to perceive at night.